Hoang Anh (Ben) NGUYEN

Email: hoanganh_nguyen@mines.edu | Personal Website: hnguyenx.com | Google Scholar

Research Summary

I am a PhD student in Geophysics at the Colorado School of Mines. My research focuses on quantum computing algorithms for solving PDEs and inverse problems. I also work on large-scale seismic multi-component elastic full waveform inversion (FWI) with GPU-accelerated solvers, as well as scientific computing and machine learning applications for geophysics.

Education

PhD in Geophysics | **Major: Geophysics, Minor: Computer Science** — Aug 2023 – May 2027 (expt.)
Colorado School of Mines (CSM) — Golden, CO, USA
Thesis: Quantum algorithms for partial differential equations and inversion

Postgraduate Diploma in Earth System Physics | **Major: Solid Earth** — Sept 2022 – Aug 2023
International Centre for Theoretical Physics (ICTP) - UNESCO — Trieste, Italy
Thesis: Ambient noise tomography beneath the Banda Arc

Master of Engineering Physics | **Major: Computational Physics** — May 2021 – Sept 2023
Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
Thesis: Structural simulation of $MgSiO_3$ under compression

Erasmus Exchange | **Major: Physics of Complex Systems** — Feb 2022 – Jul 2022
Polytechnic University of Turin (POLITO) — Torino, Italy

Bachelor of Engineering Physics | **Major: Computational Physics** — Sept 2016 – April 2021
Talent Honours Program – HUST — Hanoi, Vietnam
Thesis: Computational modeling of microstructure of magnesium silicate

Research Experience

Research Assistant — Aug 2023 – Present
Department of Geophysics, CSM — Advisor: Prof. Tura, A.

Project: End-to-end quantum algorithms for hyperbolic PDEs

- Develop explicit-circuit quantum algorithms for acoustic and elastic wave propagation based on Hamiltonian simulation and Schrödingerisation, with convolutional perfectly matched layer (C-PML) absorbing boundaries.
- Extend the formulation to the elastic velocity–stress system on staggered grids. Reflection levels reach order 10^{-4}, on par with classical C-PML implementations.

Project: Gate-based quantum computing for the forward modeling and inverse problems

- Design a hybrid quantum–classical finite-basis physics-informed neural networks (PINNs) for FWI, in which the domain-decomposed wavefield networks and the global velocity network terminate in parameterized quantum circuits.
- Implement the circuits as differentiable JAX statevector simulators. Automatic differentiation then runs end to end through the network, the circuit, and the physics-informed loss, so no parameter-shift rules are needed.
- Reach lower velocity error than the classical baseline in roughly 8× fewer training iterations, with about 33% fewer trainable parameters. The hybrid model outperforms all other classical hyperparameter variants tested.

Project: Optimization with quantum annealing

- Reformulate regularized least-squares problems as quadratic unconstrained binary optimization (QUBO) and solve them on the D-Wave Advantage annealer. Large systems are split into independent sub-problems that fit the hardware qubit limits.
- Benchmark solution accuracy and sensitivity to data noise against classical Tikhonov-regularized least squares.

Project: GPU-accelerated multi-component elastic FWI

- Lead the FWI team across code development, research directions, and results delivery.

- Build `xFWI`, a GPU-accelerated multi-component elastic FWI framework in Devito for multi-component geophone and distributed acoustic sensing data, with perfectly matched layer absorbing boundaries. The GPU implementation runs up to 100× faster than a baseline multi-core CPU version.
- Implement wavefield checkpointing for the adjoint-state gradient. Memory use stays bounded, so models larger than GPU memory can still be inverted.

Project: Time-lapse FWI for CO_2 sequestration monitoring

- Apply time-lapse FWI to baseline and monitor distributed acoustic sensing surveys to image CO_2 plume evolution and quantify time-lapse velocity and density changes.

Project: Spectral element modeling and FWI with automatic differentiation

- Develop `torchem`, a PyTorch spectral-element wave propagation framework for acoustic, elastic, SH, and coupled fluid–solid media, with anisotropy, constant-Q attenuation, and C-PML absorbing boundaries. All kernels run on CPU and GPU, and are validated against analytical solutions and SPECFEM to cross-correlation above 0.9999.
- Keep the solver end-to-end differentiable so that FWI gradients follow from automatic differentiation through the time-stepping loop, without hand-derived adjoint kernels.

Research Assistant May 2023 – Aug 2023
Earth System Physics, ICTP Advisors: Prof. Aoudia, A. & Dr. Manu-Marfo, D.

Project: Ambient noise tomography beneath the Banda arc–continent collision zone

- Developed MPI-parallelized ambient noise tomography workflows on four years of continuous seismic data. Extracted empirical Green's functions, inverted surface-wave dispersion for shear-wave velocity down to 200 km depth, and assessed model resolution.

Research Assistant Aug 2018 – July 2023
Department of Computational Physics, HUST Advisor: Prof. Nguyen, V.H.

Project: Molecular dynamics of silicate, germanate, and phosphate glasses under compression

- Developed parallel molecular dynamics codes in C on RIKEN's Fugaku supercomputer to simulate silicate, germanate, and phosphate glasses and liquids under compression. Characterized pressure-induced coordination and topological transitions.

Project: First-principles modeling of hydrogen storage alloys

- Performed first-principles density functional theory calculations with Quantum ESPRESSO. Evaluated electronic structure and formation enthalpies using plane-wave pseudopotential methods.

Work Experience

Geophysics R&D Research Intern May 2026 – Aug 2026
Shell: global integrated energy company Houston, TX

Project: Wave propagation and FWI across undulating interfaces

- Reproduced and extended the fractional grid method so that interfaces lying between grid nodes sit at their true depth in finite-difference modeling on coarse grid models.
- Built spectral-element solutions on interface-conforming curved meshes.
- Ran acoustic and elastic FWI on a high-performance computing (HPC) cluster to test how the sub-cell model-building recipe changes the inversion, from data generation on a fine reference model through inversion of coarse-grid start models.

Geophysics R&D Research Intern May 2025 – Aug 2025
TGS: energy data and analytics company Houston, TX

Project: GPU-accelerated Q-attenuation modeling and imaging in anisotropic media

- Developed GPU-accelerated seismic wave solvers for high-performance computing environments.
- Modeled Q-attenuation and compensation in isotropic and anisotropic (VTI/TTI) media using Devito Pro.

- Formulated an anisotropic attenuation model on fully staggered grids.
- Implemented Q-elastic gradient computation for reverse time migration (RTM) and FWI.

Geophysics R&D Research Intern Sept 2021 – Feb 2022
VPI: national energy research institute Hanoi, Vietnam

Project: 3D ray tracing and traveltime tomography

- Implemented 3D seismic ray-tracing algorithms for traveltime modeling in heterogeneous velocity models.
- Applied non-linear traveltime tomography techniques for subsurface velocity reconstruction.

Teaching

Teaching Assistant | GPGN 536: Advanced Geophysical Computing Fall 2026
Department of Geophysics, CSM Golden, CO

- Deliver selected lectures and lead hands-on sessions on computational methods for geophysical problems.

Current Work & Publications

Inverse problems and wave propagation

[1] Nguyen, H.A., Tura, A. Convolution absorbing boundaries for explicit-circuit quantum computing of the wave propagation. In preparation. arxiv.org/abs/2609.03440

[2] Nguyen, H.A., Vashisth, D., Tura, A. Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum–classical finite-basis architecture. Under review at *IEEE Transactions on Geoscience & Remote Sensing* (2026). arxiv.org/abs/2606.01110

[3] Nguyen, H.A., Tura, A. Joint multi-parameter elastic full-waveform inversion using distributed acoustic sensing and multi-component seismic data. Submitted to *Computers & Geosciences* (2026). arxiv.org/abs/2607.01649

[4] Nguyen, H.A., Manu-Marfo, D., & Aoudia, A. Ambient noise tomography beneath the Banda basin reveals new insights into the arc-continent collision zone. Under review at *Geophys. Res. Lett.* (2026). 10.22541/essoar.176790173.31128656/v1

[5] Nguyen, H.A., Tura, A. Seismic traveltime inversion with quantum annealing. *Scientific Reports* **15**, 17984 (2025). 10.1038/s41598-025-01188-8

[6] Pham, T.H.H., Doan, H.H., Ta, Q.M., Mai, T.L., Nguyen, H.A. Some results of seismic travel-time reflection tomography study. *Petrovietnam Journal* **10**, 4–16 (2021). 10.47800/PVJ.2021.10-01

Computational materials physics

[7] Plan, E.L.C.V.M., Phan, H., Nguyen, H.A. *et al.* Numerical simulation on structural and topological transitions of GeO_2 liquid under compression. *Eur. Phys. J. B* **99**, 20 (2026). 10.1140/epjb/s10051-026-01143-0

[8] Nguyen, G.T., Nguyen, H.A., Quan, N.T.M. *et al.* Interfacial electronic reconstruction and charge-transfer kinetics in colloidally assembled S-scheme $MoSe_2$/porphyrin heterostructures for solar water splitting. *Nanoscale Advances* (2026). 10.1039/D6NA00780E

[9] Lai, D.V., Nguyen, H.S., Nguyen, H.A. *et al.* Tailoring hydrogen storage performance of Mg–Mg_2Ni alloys: synergistic effects of composition and phase formation with first-principles insights. *RSC Advances* **15**, 31240–31254 (2025). 10.1039/D5RA04356E

[10] Nguyen, V.H., Nguyen, H.A. Crystallisation of liquid silica under compression: a molecular dynamics simulation. *Pramana - J Phys* **98**, 142 (2024). 10.1007/s12043-024-02839-7

[11] Nguyen, H.A., Nguyen, V.H. Study of the structure of $MgSiO_3$ system under compression by using ring statistics and Voronoi analysis. *Phys. Scr.* **98**, 045919 (2023). 10.1088/1402-4896/acc5b7

[12] Nguyen, V.H., Nguyen, H.A., Iitaka, T., Mai, T.L. Computer simulation of phosphate-silicate and calcium phosphate-silicate systems. *Phys. Scr.* **98**, 065704 (2023). 10.1088/1402-4896/acd4fb

[13] Nguyen, H.A., Nguyen, H.S., Nguyen, V.H. Pressure-induced glassy networks of enstatite ($MgSiO_3$) and forsterite (Mg_2SiO_4). *VNU J. Sci. Math. - Phys.* **39**, 1 (2023). 10.25073/2588-1124/vnumap.4767

[14] Nguyen, H.S., Nguyen, H.A., Pham, H.K., Iitaka, T., Nguyen, V.H. Topology of SiO_x units and glassy network of magnesium silicate glass under densification: correlation between radial distribution function and bond angle distribution. *Modelling Simul. Mater. Sci. Eng.* **28**, 065007 (2020). 10.1088/1361-651X/ab9bb4

[15] Nguyen, H.S., Nguyen, H.A. Structural simulation of Mg_2SiO_4 under compression. *VNU J. Sci. Math. - Phys.* **36**, 4 (2020). 10.25073/2588-1124/vnumap.4484

International Talks and Presentations

[1] IMAGE26 (oral): Nguyen, H.A., Tura, A. Full waveform inversion with hybrid quantum finite-basis physics-informed neural network — Aug 2026, Houston, TX

[2] EAGE26 (workshop): Tura, A., Yermakov, R., Nguyen, H.A., Sabbagh, R. Industry value of PS-wave data for Exploration and Production — Jun 2026, Aberdeen, Scotland

[3] IAGA-IASPEI25 (poster): Manu-Marfo, D., Nguyen, H.A., Aoudia, A. Ambient noise tomography beneath the Banda basin reveals new insights into the arc-continent collision zone — Sep 2025, Lisbon, Portugal

[4] IMAGE25 (workshop): Nguyen, H.A. Optimization with quantum annealing method — Aug 2025, Houston, TX

[5] IMAGE25 (oral): Nguyen, H.A., Tura, A. Crosswell traveltime inversion using a quantum computing method — Aug 2025, Houston, TX

Service and Mentoring

[1] Peer-review service

- Journal of Computational Physics (Elsevier)
- Geophysical Journal International (Oxford)
- IMAGE (SEG)
- Acta Geophysica (Springer)

[2] PhD application mentoring: mentored 4 applicants, all of whom received PhD offers:

- Department of Material Design and Innovation, University at Buffalo
- Department of Earth and Atmospheric Sciences, University of Houston
- Department of Civil and Environmental Engineering, University of Southern California
- Department of Earth and Planetary Sciences, Jackson School of Geosciences, University of Texas at Austin

Honors and Awards

[1] Denver Geophysical Society (DGS) Scholarship 2026, for research on elastic DAS FWI and quantum algorithms

[2] Fully funded scholarship for postgraduate program at ICTP 2023

[3] VEF 2.0 Program recommended candidate 2023: The VEF 2.0 Program is conducted by the Fellows and Scholars of the Vietnam Education Foundation (VEF) – an independent U.S. Federal Government agency created by the U.S. Congress

[4] Erasmus scholarship for master exchange students at POLITO 2022

[5] Fully funded scholarship for the master program at HUST 2021, 2022

[6] Certificate of Merit from School of Engineering Physics for undergraduates: Excellent Student in Fall Semester 2017, Spring Semester 2018, Fall Semester 2018, Spring Semester 2019

[7] The 20th Vietnam National Student Physics Olympiad 2018: Second Prize

[8] Lawrence S. Ting Scholarship 2017 for undergraduates

[9] Vietnam Robot National Contest 2019: team member of HUST

Certifications

[1] Deep Learning Specialization from DeepLearning.AI (Certificate)

[2] Quantum Computing and Quantum Information from IBM (Certificate)

Technical Skills

Programming: Python, Julia, C, C++, Fortran, MATLAB
Machine Learning Frameworks: JAX, PyTorch, TensorFlow, scikit-learn
Geophysical Modeling: Anisotropic elastic FWI, RTM, DAS FWI, ambient noise tomography, quantum PDE solvers
Optimization: Adjoint-state methods, quantum annealing
HPC: Slurm, MPI, OpenMP/OpenACC, GPU computing